

CMP513 - Computer Graphics and Image Processing

Tutorial Activity

Objective: Calculate precise pixel paths, update decision parameters manually, and analyze the mathematical invariants of Bresenham's algorithm.

1. Read the input pair of two end points (X_1, Y_1) and (X_2, Y_2)
Store left point in X_1, Y_1
2. Plot the first point be (X_1, Y_1)
3. Determine the initial value of the decision parameter by calculating the constants dx , dy , $2dy$ and $2(dy-dx)$ as
$$dx = |x_2 - x_1|$$
$$dy = |y_2 - y_1|$$
$$D_0 = 2dy - dx$$
4. For each X_k , conduct the following test, starting from $k = 0$
If $D_k < 0$
 Plot next point at (X_{k+1}, Y_k) and $D_{k+1} = D_k + 2dy$
Else
 Plot next point at (X_{k+1}, Y_{k+1}) and $D_{k+1} = D_k + 2dy - 2dx$
 (update $Y_k = Y_{k+1}$) (update $X_k = X_{k+1}$)
5. Repeat step (4) dx times

Problem 1 - Given coordinates are - Start Point $(X_1, Y_1) = (2, 3)$ to End Point $(X_2, Y_2) = (10, 7)$, compute the initial variables, construct a step-by-step trace table, and isolate the exact path of pixels rendered on the screen.

Calculate following terms

- $\Delta x =$ _____ ,
- $\Delta y =$ _____
- $2\Delta y =$ _____ ,
- $2\Delta y - 2\Delta x =$ _____
- Initial Decision Parameter $D_0 =$ _____

Trace table

k	D _k	x	y	Nature of D	Plot(x _{k+1} , y _{k+1})
0					
1					
2					
3					
4					
5					
6					
7					

Steep negative slope

Problem 2 - Given coordinates are - Start Point (X₁, Y₁) = (10, 12) to End Point (X₂, Y₂) = (7, 4)

Adapt the algorithm parameters to a steep, downward line where $|\Delta y| > |\Delta x|$. Show your step derivation logic and build the resulting pixel track list.

Hint: In this quadrant, your step driver shifts from stepping continuously along x to stepping continuously along y. Define your new conditional update parameter adjustments before tracking.

Visualization and graph mapping

Use layout grid (graph) paper to visually complete the tasks given below.

1. Use Blue Pen, shade the pixel boxes generated in Problem 1. Using a pencil, shade the pixel boxes generated in Problem 2.
2. Using a straightedge ruler, draw crisp vector lines linking the precise floating coordinates:
 - Line A: From exactly (2, 3) to (10, 7)
 - Line B: From exactly (10, 12) to (7, 4)

Then trace the lines using Bresenham's algorithm using pencil and highlight rejected candidate pixels with blue shade.

3. **Midpoint Observations** - Observe how the decision variable D_k structurally mirrors whether the ideal line is mathematically positioned above or below the true midpoint bounding region of the nearest physical pixels.